RECONCILIATION AGENT · SOURCE SIMULATOR · SUMMARY

How the data fits together

How each record is identified, how one item is followed from the plan to the warehouse and into Oracle, and why the agent compares item totals rather than order line against order line.

For **warehouse and operations leaders**, Lehigh Valley bakery (UBD) · November 2026 · condensed from the full 19-page guide

4	8	5 + 3	19
systems: MIPS, ODI, WMS, Oracle	record types in the simulator	5 from Bimbo reports, 3 added by us	checks run on every change

1 Each system answers a different question

MIPS plans what to make. ODI records what the plant made. WMS receives, allocates and ships it. ORACLE books the stock and the money. None of them holds the whole picture.

2 No order-line number is shared by all the systems

MIPS splits an item's demand into customer orders and forecast, and does not record customers. WMS splits the same demand by customer and has no MIPS line numbers. The two splits do not line up, so one line cannot be matched to another.

3 The item number is the common thread

The agent gathers every line for the same **item and sales day** in each system, adds them up and compares the totals. Item numbers are cleaned first because each system writes them differently. Descriptions are never used, because they differ between systems.

4 Where records do link directly, the agent follows the links

Production run → receipt ticket → warehouse movement → Oracle posting, and customer item line → customer order → shipment → Oracle posting. Where a chain breaks, for example a shipment Oracle never posts, the agent raises a flag.

5 Systems do not update each other

A change in one system stays there until a person, an interface or an approved correction changes the others. These gaps are what the worklist shows. The simulator creates them on purpose so that we can watch the agent find them.

All examples use simulated data. Item numbers run from 900101 to 900112, and every product and customer name starts with "SIM". Bimbo's own reports appear only as counts and formats.

SYSTEMS AND SOURCES

Four systems, eight record types

Plan · MIPS
How many of each item for the sales day

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Bake · ODI
Dough batches, runs, units wrapped

›

Receive · WMS
Product arrives from the plant on a ticket

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Ship · WMS
Orders are allocated, loaded and shipped

›

Book · Oracle
Every movement is posted to inventory and finance

SIMULATOR TAB	ONE ROW MEANS	CAME FROM	ROW ID	ITEM COLUMN	REAL	SIM
MIPS **Order lines**	One plan line: a customer order or a forecast	BIMBO MIPS Order Data	`ML-0001`	FG Oracle Ref	157	17
ODI **Finished goods**	One item made in one production run	BIMBO Finished Goods	`FG-0001`	OracleFGNumber	86	14
ODI **Batches**	The dough mixing for one run code	BIMBO Batch Data	`BT-0001`	none (run code)	36	6
WMS **Item lines**	One customer's order for one item	BIMBO Summary by Item	`WL-0001`	Item Nbr	364	23
WMS **Orders**	A customer's whole order and its progress	BIMBO Fill Rate Report	Order Nbr (real)	none (customer)	18	4
WMS **Tickets**	One receipt of product from a production run	ADDED no report	`TK-0001`	Item Nbr	–	11
WMS **Transactions**	One stock movement: receipt, ship, return, freezer	ADDED no report	`WT-0001`	Item Nbr	–	11
ORACLE **Postings**	Oracle's record of one warehouse movement	ADDED no report	`OT-0001`	ItemNumber	–	11

Every row ID except Order Nbr was added by us. “Real” is Bimbo's reports for Sat 12 Sep 2026: 80 items, 76 of them in MIPS, ODI and WMS, and 16 customers. “Sim” is the simulator's starting day, Sat 7 Nov 2026: 12 items, 4 customers.

What we added, and why

Row IDs (ML-, FG-, BT-, WL-), because the reports have no line numbers and a live change must say which row it changes. **Tickets, movements and Oracle postings**, because the reports stop at “shipped”, and checking that every movement reaches Oracle needs the movements themselves. Their shape is our assumption.

One item number, three spellings

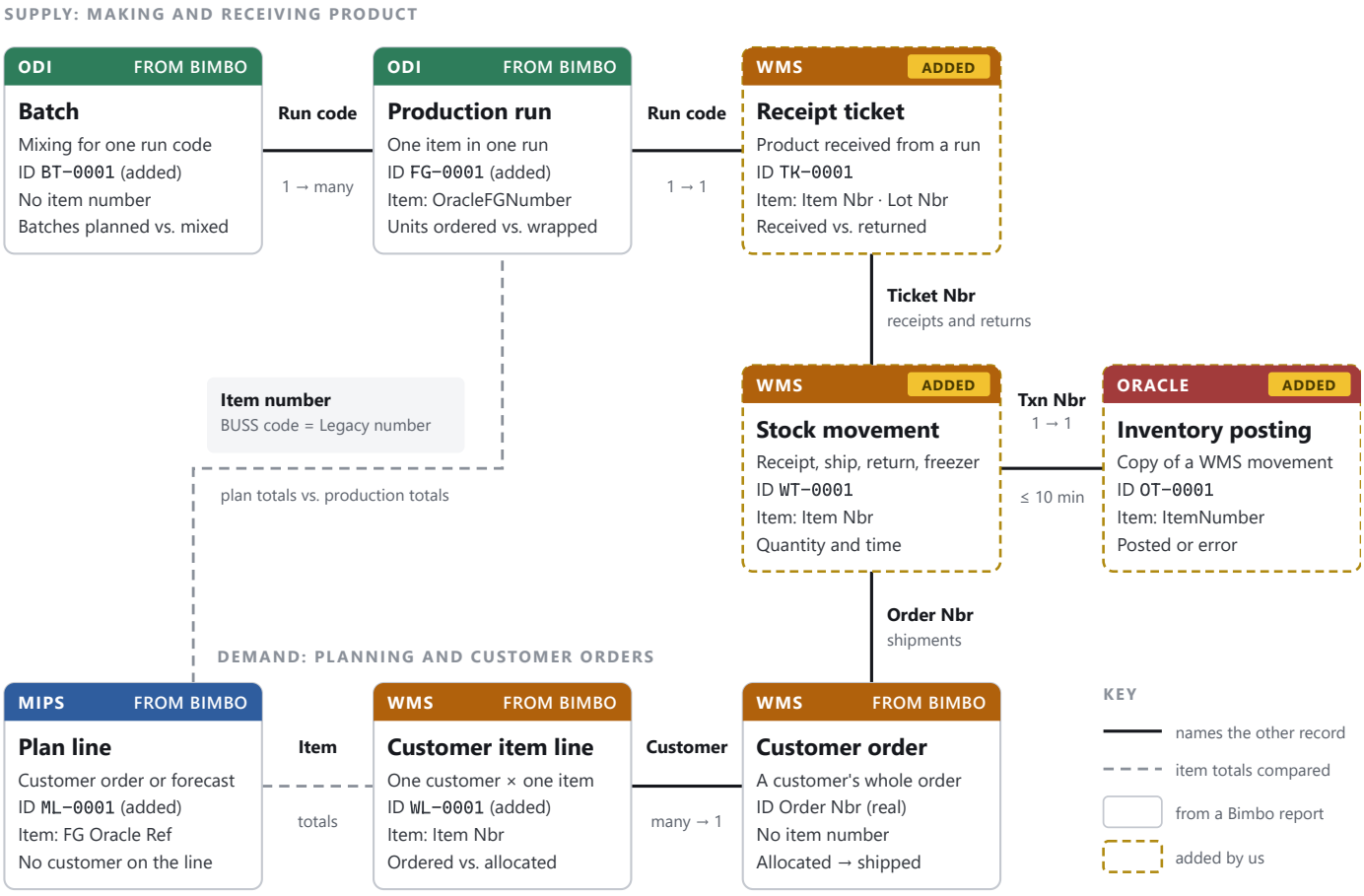
MIPS customer orders pad the number with spaces (`"900101 "`). ODI writes it as plain text. The WMS report stores it as a number. The agent cleans all three to one key before comparing. Descriptions differ between systems for about 70% of real items, so they are never used for matching.



THE DATA MAP

How the eight record types connect

A **solid line** means one record names the other directly. A **dashed line** means the records share only the item number, so the agent compares their totals per item and sales day.



- **Supply chain (top):** one batch record per run code, and one code can make several items. Each run that wraps product gets one receipt ticket, each ticket a receipt movement, and every movement should post in Oracle within 10 minutes.
- **Demand chain (bottom):** one order per customer and one item line per item ordered. Shipping creates one movement per item, and each movement names the order.
- **The MIPS plan line has no direct links.** It knows no customers, tickets or orders. It is compared with the other systems by item number and sales day only.
- **Item lines link to orders by customer name.** In the real report two customers have two orders each, so the name alone cannot say which order a line belongs to.

ONE ITEM THROUGH EVERY SYSTEM

Following item 900101 line by line

MIPS · WHAT TO MAKE

ML-0001 · customer order
BUSS · moved from Fri 11/06 — **1,300**
counts for Friday, not Saturday

ML-0002 · customer order
BUSS · Sat 11/07 — **2,400**

ML-0003 · forecast
OCloud supply plan · Sat 11/07 — **7,600**

MIPS DEMAND, ALL LINES — 11,300
for Sat 11/07: 2,400 + 7,600 = **10,000**
sent to production: **9,000** · stock used: 2,300

MADE · RECEIVED · BOOKED

BT-0001 · dough batches
run SU900110 · 12 planned, 12 mixed

run code

FG-0001 · production run
SU900110 · Line 1, Shift 1 · Wed 11/04
ordered 9,000 · wrapped — **9,000**

run code

TK-0001 · receipt ticket
received 9,000 · lot L261104-0110

ticket nbr

WT-0001 · RECEIPT
9,000 into stock · Wed 11/04 10:00

txn nbr = SourceReference

OT-0001 · Oracle posting
POSTED 2 minutes later · +9,000

WMS · WHO GETS IT

WL-0001 · SIM DC NORTH
allocated 5,000 — **5,000**
customer → order 9900001

WL-0002 · SIM DC SOUTH
allocated 3,000 — **3,000**
customer → order 9900002

WL-0003 · SIM DSD ROUTE 12
allocated 2,000 — **2,000**
customer → order 9900003

WMS ORDERS, SAT 11/07 — 10,000
allocated: **10,000**
3 customers on 3 different orders

MATCHED ON ITEM 900101 AND SALES DAY · TOTALS COMPARED

✓ Orders match plan

WMS orders for Saturday	**10,000**
MIPS demand for Saturday	**10,000**
WMS allocated	**10,000**

No R09 (order mismatch) and no R10 (allocation short). The moved 1,300 belongs to Friday, so it is correctly left out.

✓ Supply covers demand

Wrapped (ODI)	**9,000**
+ stock used by MIPS	**2,300**
= total demand, all lines	**11,300**

Nothing left to make. The 9,000 wrapped was received in full and posted to Oracle within 2 minutes.

✗ Flag R01 · High

MIPS applied the 2,300 units of stock to the **first line only** (1,300). Its requirement shows 10,000 where 9,000 is enough.

1,000 units over-production risk
$4,690 at retail
Owner: production planner

SOURCE RECORDS 3 MIPS lines · 1 ODI runs · 3 WMS orders

MIPS schedule lines

Sales day	Type	Moved	Qty	Modified
FRI 11-06	Sales order BUSS	Yes	1,300	11/05 20:31
SAT 11-07	Sales order BUSS	No	2,400	11/05 20:31
SAT 11-07	Supply plan OCloud Region2	No	7,600	11/03 15:32

ODI production runs

Run	Line · shift	Start	Ordered	Wrapped	Ending inv.
SU900110	Line 1 · Shift 1	11/04 06:00	9,000	9,000	0

WMS customer orders

Customer	Status	Ordered	Allocated
SIM DC NORTH	GREEN	5,000	5,000
SIM DC SOUTH	GREEN	3,000	3,000
SIM DSD ROUTE 12	GREEN	2,000	2,000

The worklist lists these records for the item, so a supervisor can check the evidence without opening each system.

Why not pair the lines one to one? MIPS splits Saturday's 10,000 by kind of demand: 2,400 of customer orders and 7,600 of forecast. WMS splits it by customer: 5,000, 3,000 and 2,000. Neither system stores the other's line ID. The total for each item and sales day is the only figure they share. The Friday line is an order moved onto Saturday's schedule, so it counts for Friday.

HOW A CHANGE TRAVELS

From a change in one system to a flag

Today Bimbo sends five Excel reports once a day. In the live setup, each change is sent as it happens. The simulator plays that live feed. Both routes are read by the same code, using Bimbo's own column names.

1 · Change
A row changes: an edit in the simulator, a scenario, or an approved correction

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2 · Integration
The interface carries it, and can delay, lose or duplicate it

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3 · Event log
Every change is kept in order

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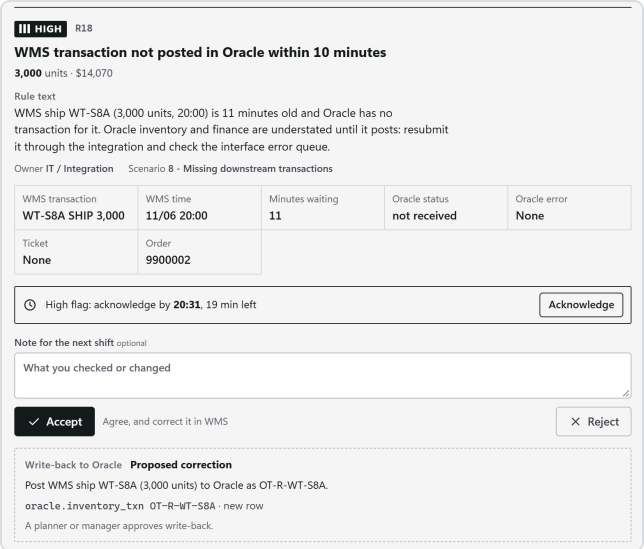
4 · Worklist
Within a second it re-adds every item and runs all 19 checks

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5 · Decision
Approved fixes go back to the source system and are confirmed when they return

Example: a shipment that never reached Oracle (scenario S8)

PLANT TIME	WHAT HAPPENS
20:00	The SIM DC SOUTH trailer ships order 9900002. WMS records six SHIP movements, one per item, each naming the order. Item 900101's 3,000 units are `WT-S8A`.
20:02	Oracle posts four of them, rejects one ("Lot not found in subinventory FG") and never receives WT-S8A.
20:10	Ten plant minutes have passed. The worklist raises **R18** (High) on both missing postings.



The R18 flag on item 900101. It names the movement, the order, the wait and Oracle's status, and proposes the fix for approval.

How it was found: each WMS movement is matched to an Oracle posting by **Txn Nbr = SourceReference**. WT-S8A has no match at all, and the rejected one is in ERROR, which counts as not posted.

> **Why it matters to the warehouse:** the product has left the building, but until Oracle posts the shipment, Oracle still counts it as stock and finance is wrong. Without the agent, gaps like this usually surface days later, at stock count or month end.

WHAT THE AGENT CHECKS

Nineteen checks, by the systems they compare

CHECK	WHAT IT CATCHES	SEVERITY
MIPS plan compared with ODI production		
R01	Stock applied to the first order line only, so MIPS asks for more than needed	High
R02	Stock taken off demand, but nothing records where it is	Medium
R03 · R04	A "blind" zero line; a moved run that left a ghost copy	Low · Medium
R05	A recovery run made far more than the shortfall	High / Medium
R06 · R08	Surplus not carried forward; stock already exceeds demand	Medium
R16	ODI's copy of the MIPS order quantity is out of date	Medium
MIPS plan compared with WMS warehouse		
R07	WMS shipped stock MIPS cannot see, often freezer stock	High / Medium
R09 · R10	Order quantities disagree; WMS allocated less than ordered	Medium
R11	MIPS and WMS disagree on units per tray	Medium
Inside WMS, ODI mixing, and data quality		
R12	Orders shipped with no trailer load scan	High
R13	Order headers and item lines do not add up	Medium
R14	More or fewer dough batches mixed than planned	Low
R15 · R17	Item numbers formatted differently; a value far outside the normal range	Low · Medium
WMS compared with ORACLE (simulation only until the live feed exists)		
R18	A movement not posted in Oracle within 10 minutes	High
R19	A return to the plant not taken off the received quantity	High / Medium

Flags are ranked by dollars at risk × severity × how reliable the check has proved. High flags must be acknowledged within 15 plant minutes.

What Bimbo's real reports showed (sample day 12 Sep 2026, counts only). The real data raised 104 flags from 16 checks. For the warehouse:

- **Loaded Qty is 0 on every order** in the Fill Rate report, while Packed, Allocated and Shipped are equal. Either trailer load scans are not captured, or the report does not show them.
- **Status is GREEN on every order**, including 5 that are short. The color cannot be relied on to show a shortfall.
- **Order headers and item lines disagree for 4 customers.** Units per container differ between MIPS and WMS for 3 items.

WHAT WE NEED TO CONFIRM

Assumptions and open questions

The simulator rests on these assumptions about the real systems. If one proves wrong, the matching described here changes with it.

#	WE ASSUMED	WHY IT MATTERS
A1–A3	MIPS plan lines, ODI runs and ODI batches each have a permanent ID, and MIPS keeps an item's lines in a fixed order	A live change must say which row it changes. R01 depends on which line is first
A4	WMS provides order headers and order lines separately, ideally with the order number on each line	Today lines link to orders only by customer name
A5–A7	Each change arrives with a timestamp, in order, and as the complete row; column names match today's reports	Needed to know how current the worklist is, and to repair a lost change
A8, A10	Oracle postings carry the WMS transaction number and a posted or error status; the 10-minute limit runs from the WMS movement time	R18 depends on both
A9	WMS tickets keep a returned quantity, and each return names its ticket	R19 compares the two
A11	Oracle Integration accepts a correction as a row change and checks the row has not changed in the meantime	Needed to send approved fixes back safely

Questions for the warehouse team

- Are trailer load scans recorded somewhere the Fill Rate report does not show?
- What turns an order or item line from GREEN to YELLOW in WMS?
- How are receipts, returns to the plant and freezer withdrawals recorded today, and what reference reaches Oracle with each?
- When a customer has two orders on one sales day, how is an item line tied to the right order?

Key terms

BUSS / OCloud	The two kinds of MIPS plan line: customer sales orders, and forecast (supply-planning) lines
Sales day	The day the product is sold. Plan lines, orders and flags all belong to a sales day
Moved line	A MIPS line on today's schedule whose demand belongs to an earlier sales day
Run code	The run identifier (e.g. SU900110) shared by runs, batches and tickets. A code ending in “A” is a moved run
Ticket · movement · posting	A WMS receipt from a run · a WMS stock change · Oracle's record of that change
Write-back	A fix sent from the worklist to a source system after a planner or manager approves it

Full detail, including every record type with screenshots, is in **Data_Flow_and_Mapping_Guide.pdf**.